Tanvir Hasan

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Research Interests

- Distributed Computing
- Machine Learning
- Cloud Computing
- Deep Learning
- Signal Processing
- Software Engineering

Education

B.Sc.(Engg.) in Computer Science & Engineering Feb 2017 - Oct 2022
Chittagong University of Engineering & Technology (CUET)
CGPA: 2.90/4.0
Academic Thesis: A Comparative Study of Machine Learning and Deep Learning Models for EEG-Based Seizure Detection.

- Implemented and evaluated CNN, RNN, LSTM architectures and classical ML classifiers on EEG seizure detection datasets, achieving high detection accuracy while conducting comprehensive performance -complexity trade-off analysis
- Conducted systematic comparative analysis across multiple models, providing insights into model selection for real-time biomedical signal processing applications

Standardized Tests

Test	Score	Date
GRE General Test	Q: 161 | V: 138 | AWA: 2.5	11 August 2025
IELTS Academic	Scheduled for March 2026	—

Professional Experience

Telcobright Ltd. **Dec 2022 – Present**
Software Engineer

- Architected and implemented a fault-tolerant **distributed stream processing system** from scratch for real-time data ingestion and processing at scale, applying distributed systems principles including consensus algorithms, state management, and exactly-once semantics
- Designed and maintained **microservices-based** backend systems using Java, Spring Boot, and Quarkus, implementing REST APIs, gRPC services, and JPA/Hibernate for data persistence with focus on scalability and performance optimization
- Established comprehensive security infrastructure with **Spring Security** and API Gateway, implementing authentication, authorization, and API routing patterns to ensure secure inter-service communication in distributed environments
- Deployed production-grade systems on cloud infrastructure using AWS EC2, Docker containerization, database replication strategies, and automated CI/CD pipelines with Terraform and Ansible, optimizing system performance and maintaining high availability
- Applied advanced software design patterns, distributed systems best practices, and performance optimization techniques to deliver scalable, maintainable, and production-ready systems across multiple critical projects

Technology Used: Java, Spring Boot, Quarkus, Kafka, Apache Flink, gRPC, Docker, JPA/Hibernate, AWS EC2, Terraform, Ansible, CI/CD

Relevant Coursework

Algorithm Design, Data Structures, Operating Systems, Compiler Design, Machine Learning, Software Engineering, Networking, Artificial Intelligence, Digital Signal Processing, Digital Systems Design

Computing Skills

- **Programming Language:** C/C++, Java, C#, Python
- **Databases & Messaging:** MySQL, MariaDB, Redis, Apache Kafka
- **Version Control:** Git
- **Cloud & Infrastructure:** AWS, Terraform, Jenkins (CI/CD).
- **Container Technologies:** Docker,LXD
- **Operating Systems:** Linux kernel-based distributions (Ubuntu, Debian, CentOS etc.), Windows

Projects

Multi-Tenant Microservices Framework GitHub Repository

- Developing **production-ready microservices framework** with **Gateway**, **Service Discovery**, **Config Server**, JWT authentication, and RBAC
- Enabling **modular microservices integration** for scalable deployment with **per-tenant resource allocation** and isolation

Real-Time Distributed Messaging System GitHub Repository

- Built high-performance server using **multithreading, WebSocket** for real-time communication, streaming to **Kafka** topics
- Implemented fault-tolerant pipeline with **Apache Flink**, achieving **exactly-once delivery** via checkpointing and state management

FreeSWITCH Event Socket Integration (ESL Client) GitHub Repository

- Developed **Spring Boot** microservice integrating with **FreeSWITCH** via **ESL protocol** for real-time call routing and monitoring
- Implemented asynchronous event-driven architecture for **inbound/outbound** calls with error recovery and Docker deployment

Achievements

Machine Learning Specialization - Stanford University Online (Coursera)
Instructed by Prof. Andrew Ng

- Completed comprehensive study of supervised and unsupervised learning algorithms, including linear/logistic regression, neural networks, support vector machines, decision trees, clustering algorithms, and dimensionality reduction techniques
- Gained hands-on implementation experience with TensorFlow and NumPy, applying ML concepts to real-world datasets

Competitive Programming

- Achieved **Pupil rating (1374)** on Codeforces, demonstrating strong problem-solving and algorithmic thinking skills (Handle: Veer)
- Solved **1500+ algorithmic problems** across multiple online judges (Codeforces, LeetCode, UVa), covering data structures, dynamic programming, graph algorithms, and computational geometry

Leadership & Service

- **Finance Secretary,** CUET CSE Fest - CUET Computer Club
  Managed financial operations, budget allocation, and fiscal planning for the annual technical festival, ensuring successful execution of the multi-day event with 500+ participants
- **Volunteer,** National Collegiate Programming Contest (NCPC) 2017
  Contributed to event organization, technical setup, and participant support for Bangladesh's premier collegiate programming competition